

Problem Statement -

Build your own recommendation system for products on an e-commerce website like Amazon.com.

Dataset columns - first three columns are userId, productId, and ratings and the fourth column is timestamp. You can discard the timestamp column as in this case you may not need to use it.

Source - Amazon Reviews data (<http://jmcauley.ucsd.edu/data/amazon/>). The repository has several datasets. For this case study, we are using the Electronics dataset.

Please do the analysis based on steps (1 to 8) as given below -

Steps -

1. Read and explore the given dataset. (Rename column/add headers, plot histograms, find data characteristics) - 5
2. Take a subset of the dataset to make it less sparse/ denser. (For example, keep the users only who has given 50 or more number of ratings.) - 10
3. Split the data randomly into train and test dataset. (For example, split it in 70/30 ratio)
4. Build Popularity Recommender model. - 10
5. Build Collaborative Filtering model. - 10
6. Evaluate both the models. (Once the model is trained on the training data, it can be used to compute the error (RMSE) on predictions made on the test data.) - 10
7. Get top - K (K = 5) recommendations. Since our goal is to recommend new products to each user based on his/her habits, we will recommend 5 new products. - 5
8. Summarize your insights. - 10

Reference links - <https://www.computer.org/csdl/mags/ic/2017/03/mic2017030012.html>